

Paper 3

Title

AI-Powered Decision Support System for Real-Time Fish Mortality Prediction Using Random Forest

Abstract

Unexpected fish mortality is a major challenge in commercial mariculture because it causes severe economic losses and complicates daily farm management. This study presents an AI-powered Decision Support System (DSS) that predicts fish mortality risk in real time using the Random Forest machine learning algorithm. Water quality information, including seawater temperature, salinity, conductivity, chlorophyll-a, turbidity, dissolved oxygen, and water current, was continuously collected through an IoT sensor network. These environmental measurements were combined with daily fish mortality records provided by farmers to construct a five-level mortality risk prediction model. Sensor data were automatically transmitted to a cloud database, where the prediction model was updated regularly to support real-time decision making. Feature importance analysis revealed that seawater temperature, salinity, and turbidity were among the most influential factors affecting mortality risk. The developed Decision Support System provides practical recommendations for feed management, labor planning, and operational scheduling, demonstrating the potential of artificial intelligence for improving aquaculture management.

Introduction

Mariculture has become an economically important industry worldwide, yet high fish mortality remains one of its most serious operational challenges. Unexpected mortality events reduce productivity, increase financial losses, and create difficulties in planning feeding schedules, labor allocation, and overall farm management. Conventional monitoring methods often depend on manual observation or threshold-based alarms, which provide limited predictive capability.

Recent developments in sensor networks and machine learning have created opportunities for intelligent decision-support systems capable of continuously monitoring environmental conditions and predicting mortality risk before major losses occur. In this study, the authors integrated a real-time IoT sensor network with a Random Forest classification model to estimate fish mortality risk in grouper mariculture farms.

The system continuously monitored multiple water quality parameters, including seawater temperature, dissolved oxygen, salinity, conductivity, chlorophyll-a, turbidity, and water current. Daily mortality records entered by fish farmers were combined with sensor measurements to train a machine learning model capable of classifying mortality risk into five levels. The prediction

results were automatically transmitted to farm managers through a cloud-based Decision Support System, enabling timely management decisions regarding feed procurement, labor scheduling, and farm operations.

Unlike earlier studies that primarily performed binary mortality prediction or retrospective analysis, this work demonstrates the feasibility of real-time mortality risk prediction integrated with operational decision support, making artificial intelligence more practical for commercial aquaculture management.